Student ID:		24CS072	Student Name:		Patel Prince			
Subject code & Name	:	CSE304 & Full Stack Development	Practical	:	7	Academic Year	:	**2025-26**

Task - 7

Aim

• Add user registration and login functionality to the Task Management application.
• Hash passwords using bcrypt before saving the User document to MongoDB.
• Implement JWT token generation on successful login, with a reasonable expiry (e.g., 1 hour).
• Protect all task routes using an authentication middleware that verifies the JWT.
• Add server-side input validation that rejects malformed requests (e.g., missing title) before they reach the database.

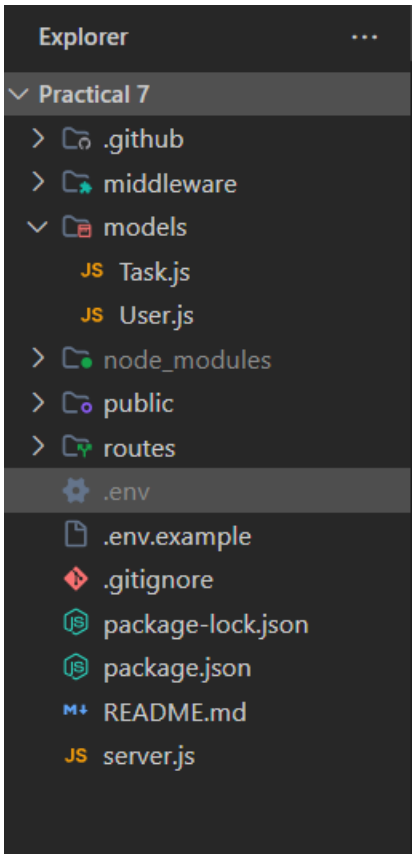
Code

GitHub Link Repository Link:

Code repo :-
https://github.com/PRINCE-24CS072/AWDF/tree/main/Practical-7

Output

Folder Structure

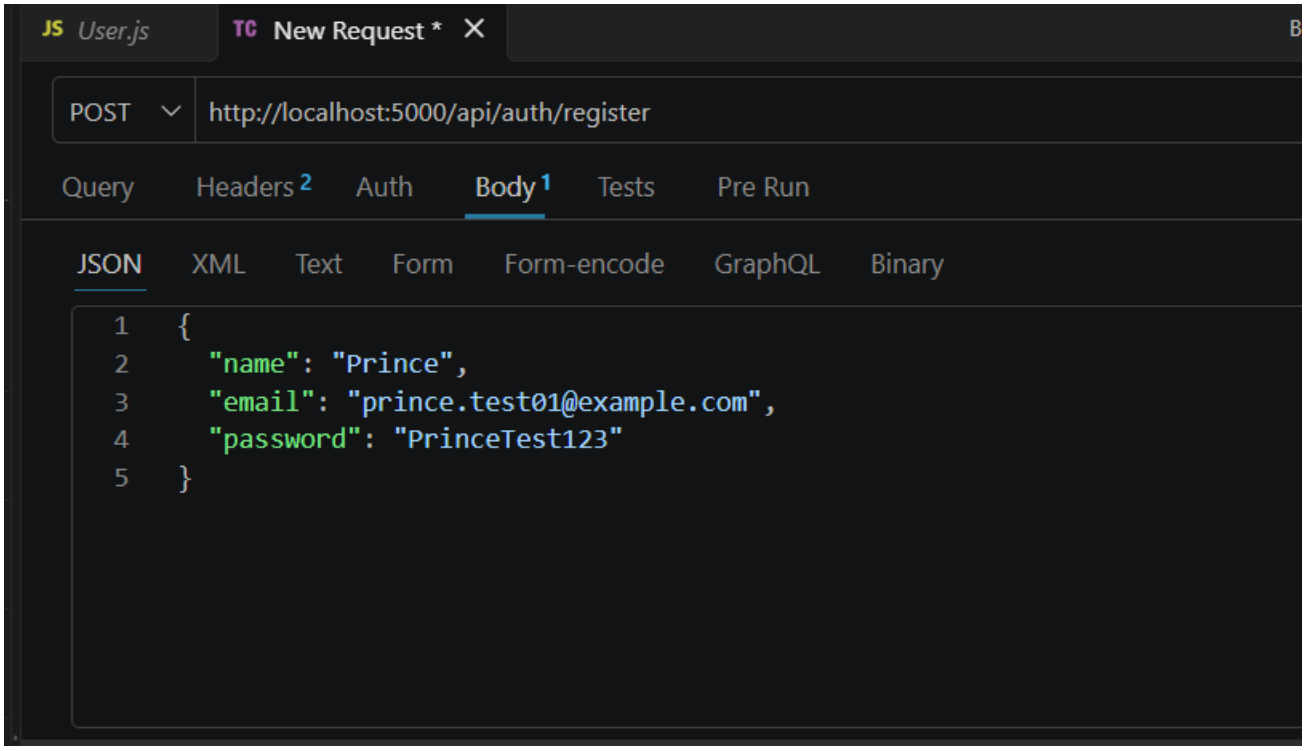


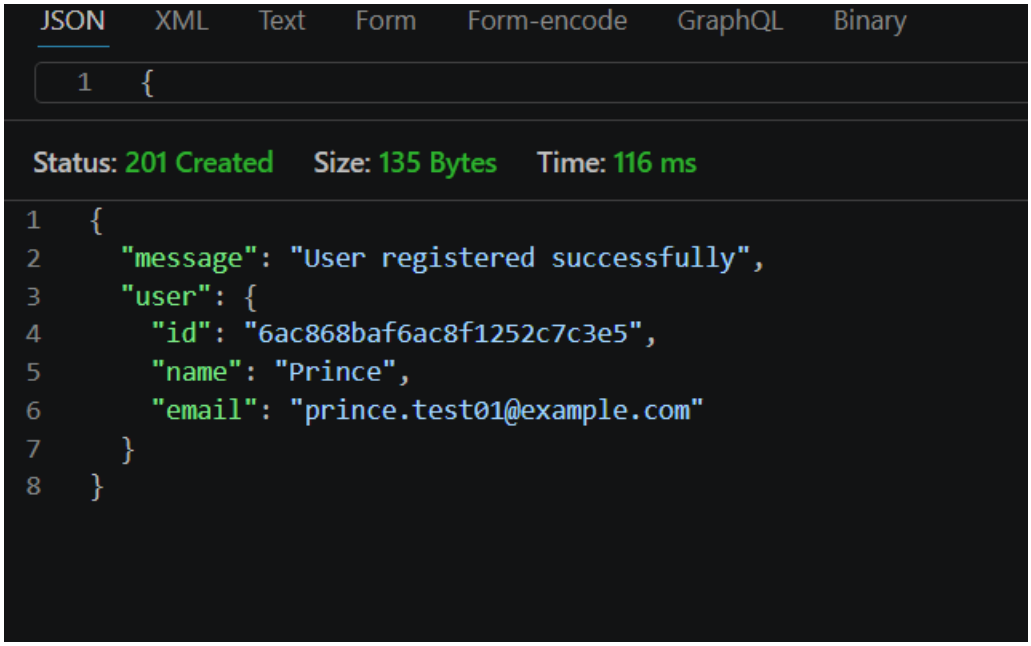
package.json

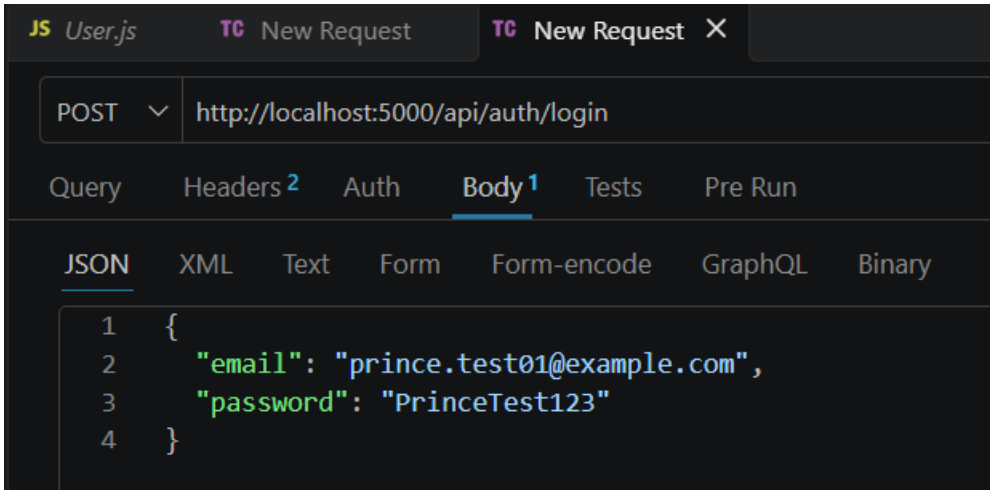
```
{
  "name": "practical-7-task-management",
  "version": "1.0.0",
  "description": "Task Management Application with JWT Authentication and Middleware Pipeline",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  },
  "dependencies": {
    "bcryptjs": "^2.4.3",
    "cors": "^2.8.5",
    "dotenv": "^16.4.5",
    "express": "^4.21.0",
    "express-validator": "^7.2.0",
    "jsonwebtoken": "^9.0.2",
    "mongoose": "^8.6.0"
  },
  "devDependencies": {
    "nodemon": "^3.1.4"
  }
}
```

.env

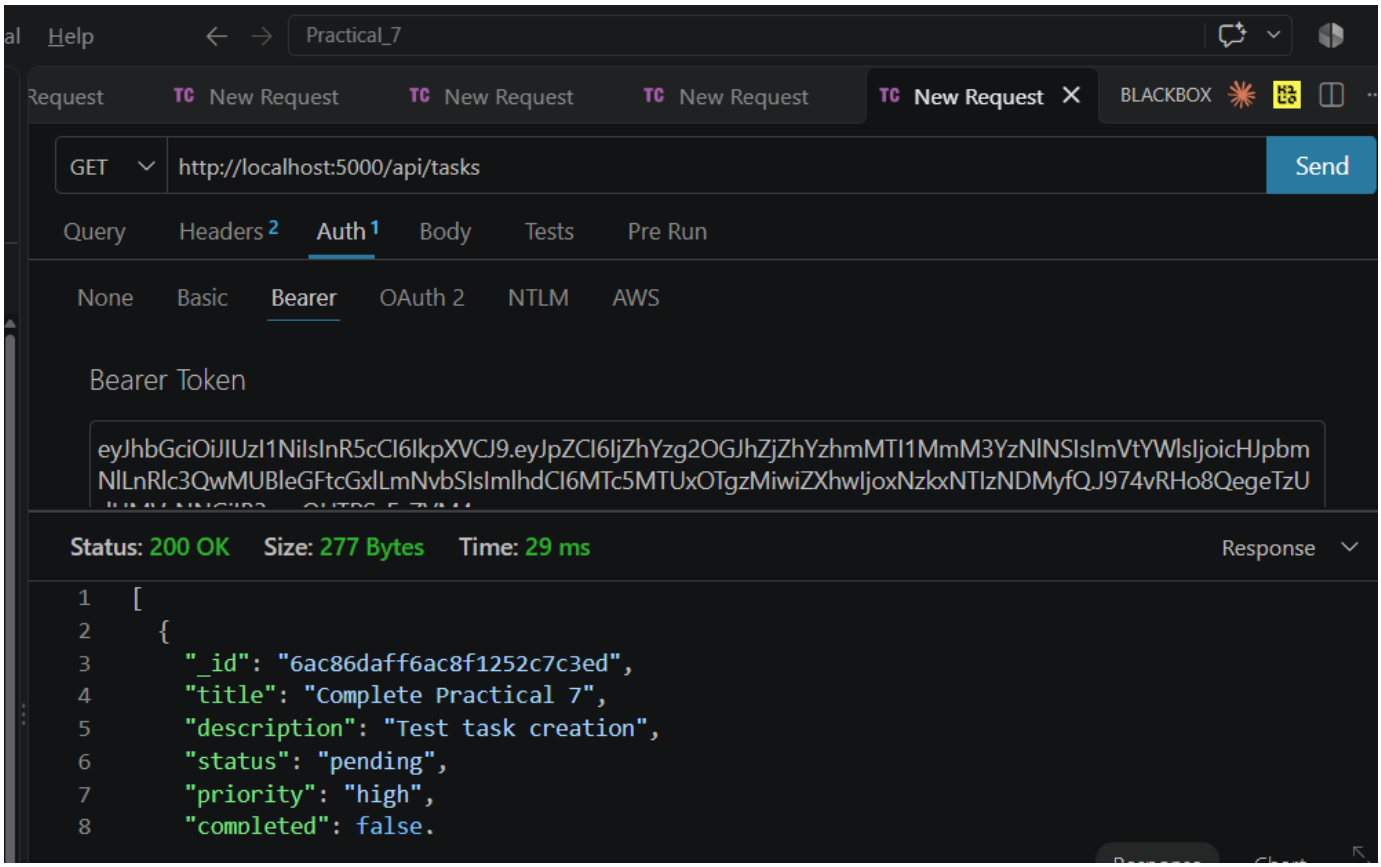
```
PORT=5000
MONGO_URI=mongodb://127.0.0.1:27017/awdf_practical7
JWT_SECRET=awdf_practical_7_super_secret_jwt_key_24cs072
```

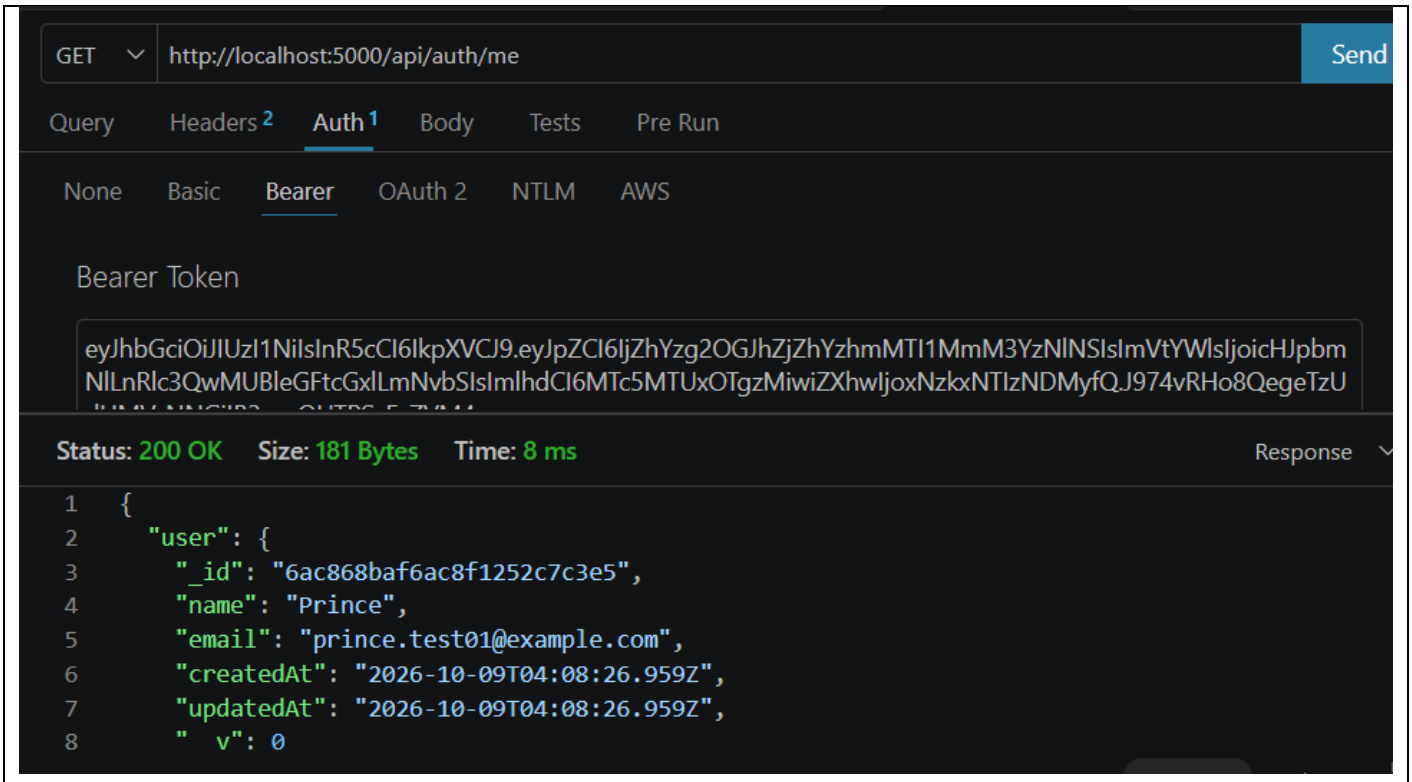

User.js
New Request *
POST
http://localhost:5000/api/auth/register
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
{
"name": "Prince",
"email": "prince.test01@example.com",
"password": "PrinceTest123"
}

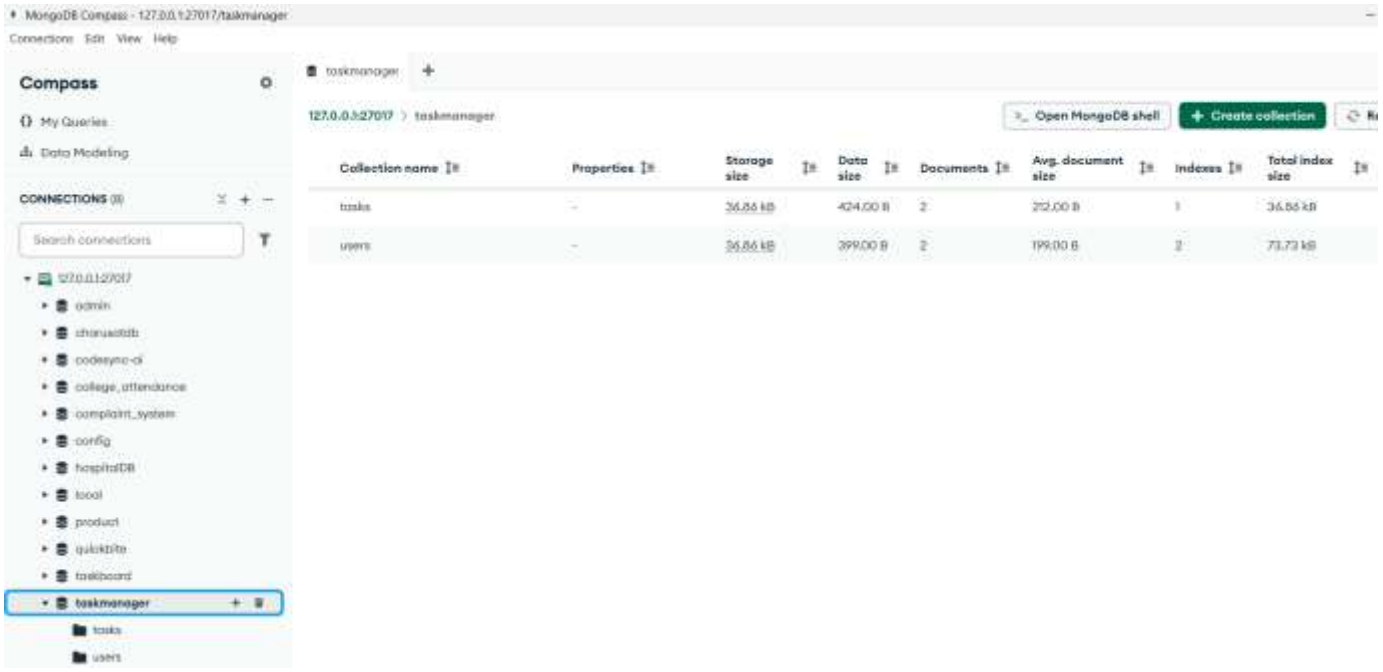

JSON
XML
Text
Form
Form-encode
GraphQL
Binary
Status: 201 Created
Size: 135 Bytes
Time: 116 ms
{
"message": "User registered successfully",
"user": {
"id": "6ac868baf6ac8f1252c7c3e5",
"name": "Prince",
"email": "prince.test01@example.com"
}
}

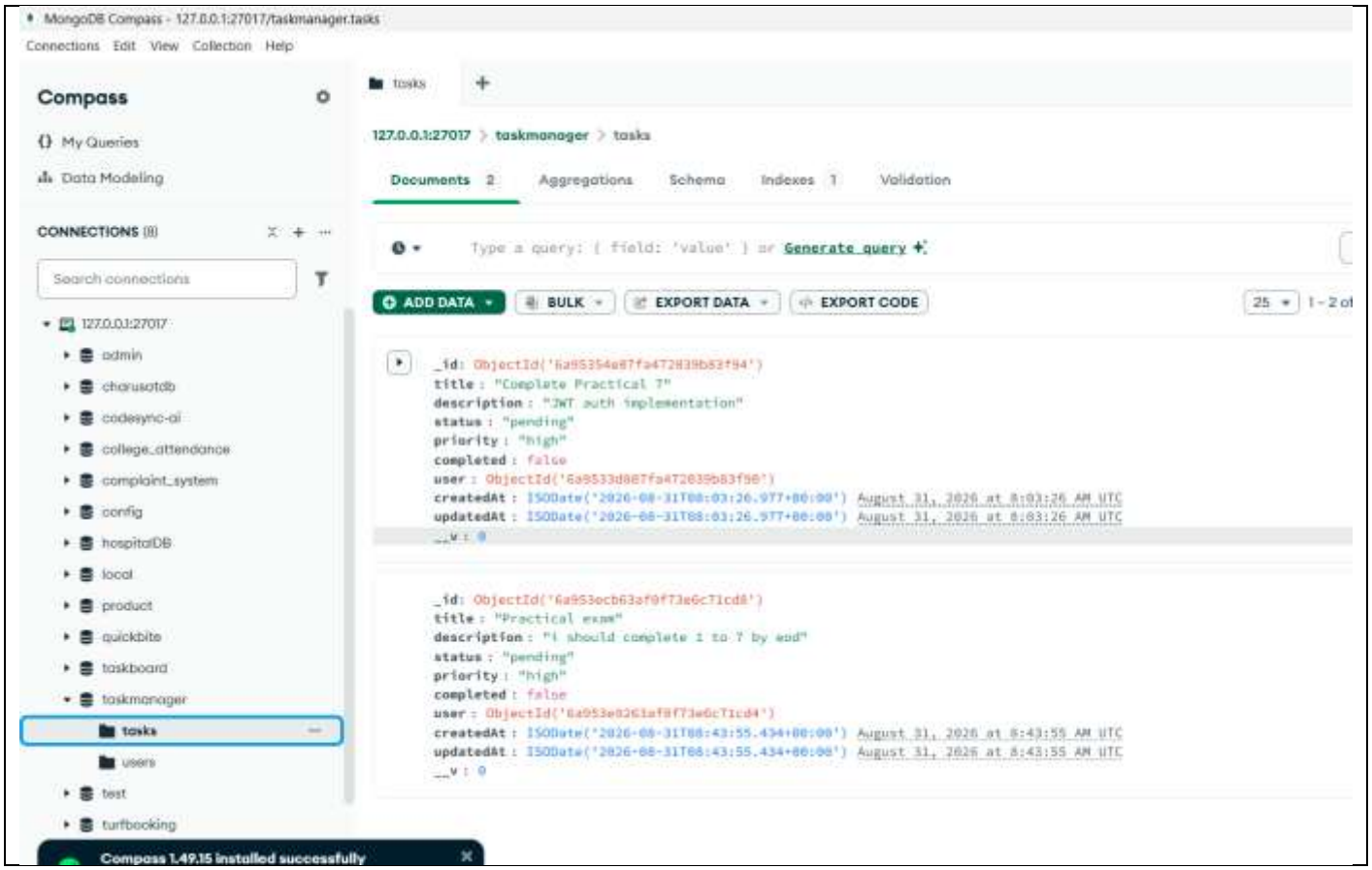

User.js
New Request
New Request
POST
http://localhost:5000/api/auth/login
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
{
"email": "prince.test01@example.com",
"password": "PrinceTest123"
}

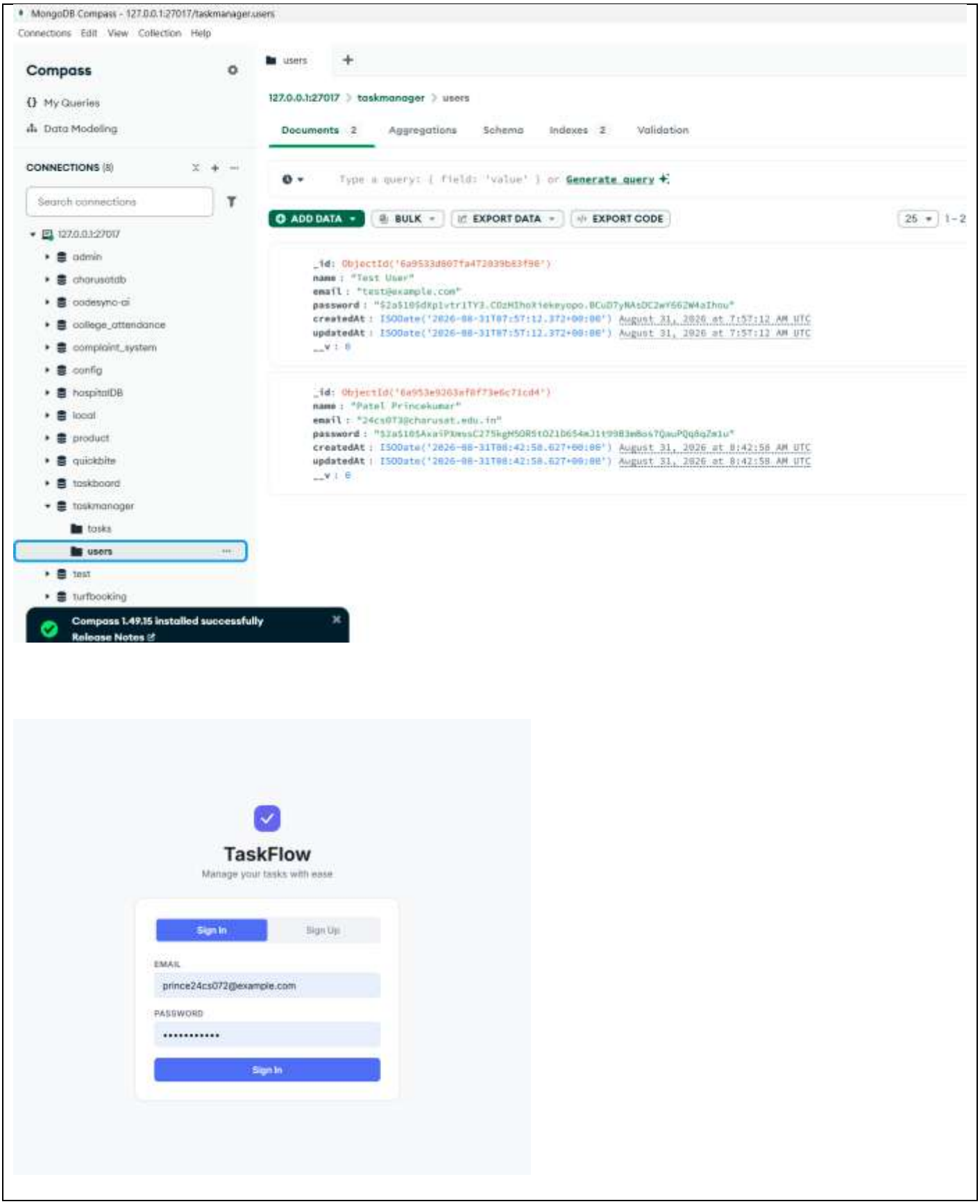

JSON
XML
Text
Form
Form-encode
GraphQL
Binary
Status: 200 OK
Size: 353 Bytes
Time: 118 ms
Response
{
"message": "Login successful",
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9
.eyJpZCI6IjZhYzg2OGJhZjZhYzhmMTI1MmM3YzNlNSIsImVtYWlsIjoicHJpbmNlLnRlc3QwMUBleGFtcGxlLmNvb
SIsImlhdCI6MTc5MTUxOTIwMywiZXhwIjoxNzkxNTIyODAzfQ.aGfIA5bkKYHF
-x6ukySKtEpvJ6A3u6o7BG24qVcvuvc",
"user": {
"id": "6ac868baf6ac8f1252c7c3e5",
"name": "Prince",
"email": "prince.test01@example.com"
}

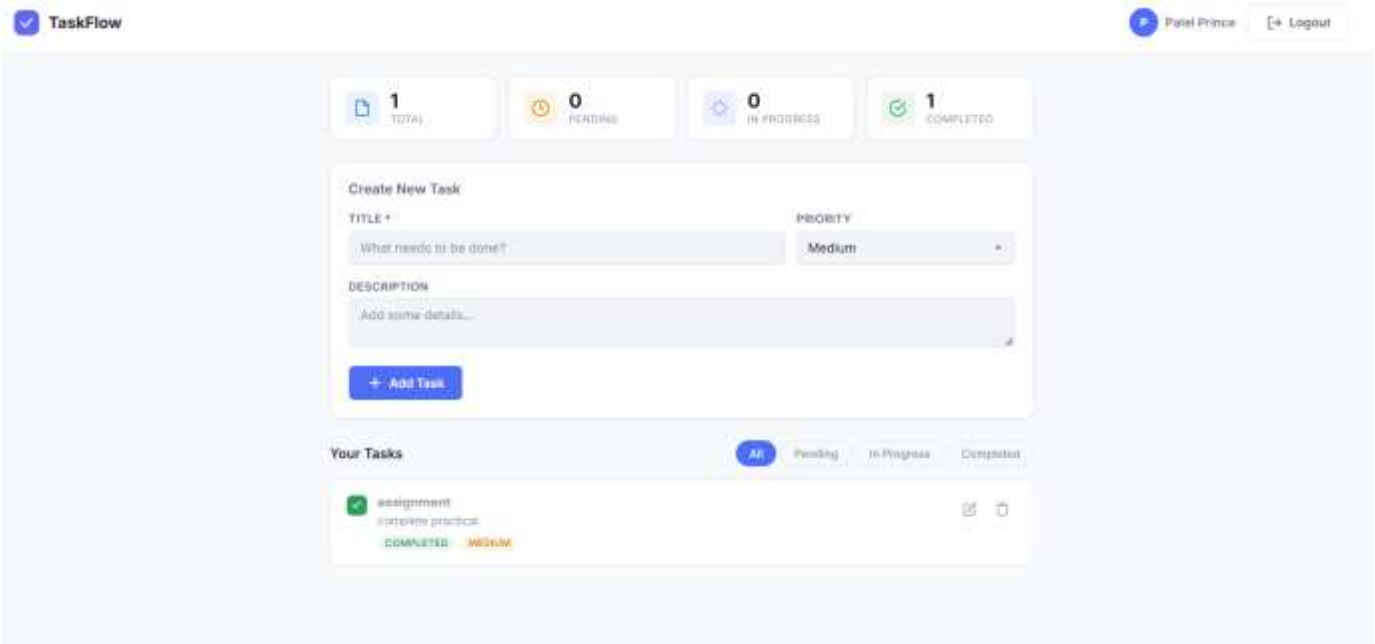

Help
Practical_7
New Request
New Request
New Request
New Request
BLACKBOX
GET
http://localhost:5000/api/tasks
Send
Query
Headers 2
Auth 1
Body
Tests
Pre Run
None
Basic
Bearer
OAuth 2
NTLM
AWS
Bearer Token
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjZhYzg2OGJhZjZhYzhmMTI1MmM3YzNlNSIsImVtYWlsIjoicHJpbmNlLnRlc3QwMUBleGFtcGxlLmNvbSIsImlhdCI6MTc5MTUxOTgzMiwiZXhwIjoxNzkxNTIzNDMyfQ.J974vRHo8QegeTzU
Status: 200 OK
Size: 277 Bytes
Time: 29 ms
Response
[
{
"_id": "6ac86daff6ac8f1252c7c3ed",
"title": "Complete Practical 7",
"description": "Test task creation",
"status": "pending",
"priority": "high",
"completed": false.


GET
http://localhost:5000/api/auth/me
Send
Query
Headers 2
Auth 1
Body
Tests
Pre Run
None
Basic
Bearer
OAuth 2
NTLM
AWS
Bearer Token
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjZhYzg2OGJhZjZhYzhmMTI1MmM3YzNlNSIsImVtYWlsIjoicHJpbmNlLnRlc3QwMUBleGFtcGxlLmNvbSIsImlhdCI6MTc5MTUxOTgzMiwiZXhwIjoxNzkxNTIzNDMyfQ.J974vRHo8QegeTzU
Status: 200 OK
Size: 181 Bytes
Time: 8 ms
Response
{
"user": {
"_id": "6ac868baf6ac8f1252c7c3e5",
"name": "Prince",
"email": "prince.test01@example.com",
"createdAt": "2026-10-09T04:08:26.959Z",
"updatedAt": "2026-10-09T04:08:26.959Z",
" v": 0


Compass
My Queries
Data Modeling
CONNECTIONS
taskmanager
tasks
users
Open MongoDB shell
Create collection
Collection name
Properties
Storage size
Data size
Documents
Avg. document size
Indexes
Total index size


MongoDB Compass - 127.0.0.1:27017/taskmanager.tasks
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (8)
Search connections
127.0.0.1:27017
admin
charusatdb
codesync-ai
college_attendance
complaint_system
config
hospitalDB
local
product
quickbite
taskboard
taskmanager
tasks
users
test
turfbooking
Compass 1.49.15 installed successfully
tasks
127.0.0.1:27017 > taskmanager > tasks
Documents 2
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 2 of
_id: ObjectId('6a9533d867fa472039b83f96')
title : "Complete Practical 7"
description : "JWT auth implementation"
status : "pending"
priority : "high"
completed : false
user : ObjectId('6a9533d867fa472039b83f96')
createdAt : ISODate('2026-08-31T08:03:26.977+00:00') August 31, 2026 at 8:03:26 AM UTC
updatedAt : ISODate('2026-08-31T08:03:26.977+00:00') August 31, 2026 at 8:03:26 AM UTC
__v : 0
_id: ObjectId('6a953ecb63af0f73e6c71cd8')
title : "Practical exam"
description : "I should complete 1 to 7 by end"
status : "pending"
priority : "high"
completed : false
user : ObjectId('6a953e0263af0f73e6c71cd4')
createdAt : ISODate('2026-08-31T08:43:55.434+00:00') August 31, 2026 at 8:43:55 AM UTC
updatedAt : ISODate('2026-08-31T08:43:55.434+00:00') August 31, 2026 at 8:43:55 AM UTC
__v : 0


MongoDB Compass - 127.0.0.1:27017/taskmanager.users
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (8)
Search connections
127.0.0.1:27017
admin
charusatdb
codesync-ai
college_attendance
complaint_system
config
hospitalDB
local
product
quickbite
taskboard
taskmanager
tasks
users
test
turfbooking
Compass 1.49.15 installed successfully
Release Notes
users
127.0.0.1:27017 > taskmanager > users
Documents 2
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 2
_id: ObjectId('6a9533d807fa472039b83f98')
name : "Test User"
email : "test@example.com"
password : "$2a$10$dXp1vtr1TY3.CDzMIhoX1ekeyopo.BCuD7yNAsDC2wY662W4aIhou"
createdAt : ISODate('2026-08-31T07:57:12.372+00:00') August 31, 2026 at 7:57:12 AM UTC
updatedAt : ISODate('2026-08-31T07:57:12.372+00:00') August 31, 2026 at 7:57:12 AM UTC
__v : 0
_id: ObjectId('6a953e9263af8f73e6c71cd4')
name : "Patel Princekumar"
email : "24cs073@charusat.edu.in"
password : "$2a$10$AxaiPXmusC275kgM5OR5tOZ1D654mJ1t99B3m8os7QauPQq8qZm1u"
createdAt : ISODate('2026-08-31T08:42:58.627+00:00') August 31, 2026 at 8:42:58 AM UTC
updatedAt : ISODate('2026-08-31T08:42:58.627+00:00') August 31, 2026 at 8:42:58 AM UTC
__v : 0
TaskFlow
Manage your tasks with ease
Sign In
Sign Up
EMAIL
prince24cs072@example.com
PASSWORD
Sign In



Terminal

```
PS C:\Users\Prince Patel\OneDrive\Desktop\LAB Submission\AWDF> cd Practical-7
PS C:\Users\Prince Patel\OneDrive\Desktop\LAB Submission\AWDF\Practical-7> npm run dev

> practical-7-task-management@1.0.0 dev
> nodemon server.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
MongoDB connected successfully
Server running on http://localhost:5000
```